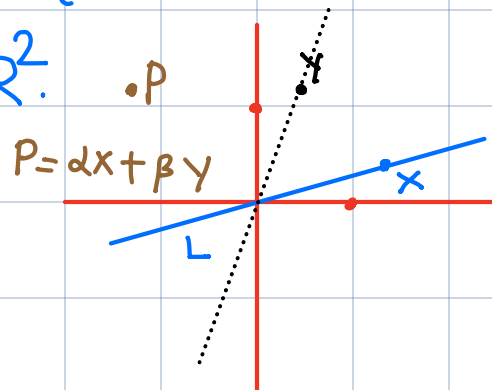


③ Let $x, y \in \mathbb{R}^2$ be two points such that $0 \neq x$, $0 \neq y$ and y is not on the line $L = \{\alpha x : \alpha \in \mathbb{R}\}$

Then $\text{Span}\{x, y\} = \mathbb{R}^2$.



Proof:

Here $\text{Span}\{x, y\} = \{\alpha x + \beta y : \alpha \in \mathbb{R}, \beta \in \mathbb{R}\}$.

We need to show

$$\mathbb{R}^2 = \{\alpha x + \beta y : \alpha \in \mathbb{R}, \beta \in \mathbb{R}\}$$

* If you need to show two sets A and B are the same then you need to show $A \subseteq B$ and $B \subseteq A$.

Subset
 $a \in A \Rightarrow a \in B$.

It is clear that $\{\alpha x + \beta y : \alpha \in \mathbb{R}, \beta \in \mathbb{R}\} \subseteq \mathbb{R}^2$

Obvious!

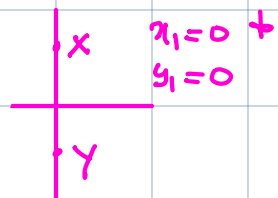
Now let us show that $\mathbb{R}^2 \subseteq \{\alpha x + \beta y : \alpha \in \mathbb{R}, \beta \in \mathbb{R}\}$

Pick any point $\begin{pmatrix} a \\ b \end{pmatrix} \in \mathbb{R}^2$.

We will show $\begin{pmatrix} a \\ b \end{pmatrix} \in \{ \alpha X + \beta Y : \alpha \in \mathbb{R}, \beta \in \mathbb{R} \}$

This will prove $\mathbb{R}^2 \subseteq \{ \alpha X + \beta Y : \alpha \in \mathbb{R}, \beta \in \mathbb{R} \}$

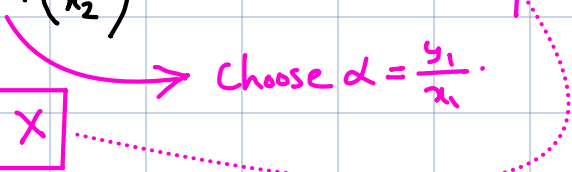
Let $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, $Y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ and $A = \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix}$.

Then $A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = X$ and $A \begin{pmatrix} 0 \\ 1 \end{pmatrix} = Y$ 

Since x_1 and y_1 both (simultaneously) cannot be zero, we may assume $x_1 \neq 0$.

We claim that $\det(A) \neq 0$. Otherwise, we have $x_1 y_2 = x_2 y_1$, which implies $y_2 = x_2 y_1 / x_1$ and we obtain

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \frac{y_1}{x_1} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \text{ a contradiction}$$

$\boxed{Y = \alpha X}$  choose $\alpha = \frac{y_1}{x_1}$.

Thus $\det(A) \neq 0$ and therefore A^{-1} exists. Now let

$$\begin{pmatrix} c \\ d \end{pmatrix} := A^{-1} \begin{pmatrix} a \\ b \end{pmatrix}$$

and observe that

$$\begin{pmatrix} c \\ d \end{pmatrix} = c \begin{pmatrix} 1 \\ 0 \end{pmatrix} + d \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow A \begin{pmatrix} c \\ d \end{pmatrix} = c A \begin{pmatrix} 1 \\ 0 \end{pmatrix} + d A \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = c X + d Y. \quad \left(\text{So } \begin{pmatrix} a \\ b \end{pmatrix} \text{ is a linear combination of } X \text{ and } Y \right)$$

This proves that $\begin{pmatrix} a \\ b \end{pmatrix} \in \left\{ \alpha X + \beta Y : \alpha, \beta \in \mathbb{R} \right\}$ $\square$

④ Let $S \subseteq V$ be a nonempty subset of a vector space V . Then $\text{Span}(S)$ is a subspace of V .

Proof: We need to check if $\text{Span}(S)$ is a vector space.

① Take two random elements from $\text{Span}(S)$, say,
 $\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n, \quad \beta_1 y_1 + \beta_2 y_2 + \dots + \beta_m y_m$

where $\alpha_1, \alpha_2, \dots, \alpha_n, \beta_1, \beta_2, \dots, \beta_m \in F$ and
 $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_m \in S$.

Now if we 'add' these two elements, we obtain

$\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n + \beta_1 y_1 + \beta_2 y_2 + \dots + \beta_m y_m,$
 which again belongs to $\text{Span}(S)$.

② Associativity holds as $\text{Span}(S) \subseteq V$.

(G3) Take $x \in S$ and $0 \in F$ and observe that
 $0 = 0 * x \in \text{Span } S$.

(G4) Inverse of $\alpha_1 x_1 + \dots + \alpha_n x_n \in \text{Span}(S)$ is
 $-\alpha_1 x_1 + (-\alpha_2) x_2 + \dots + (-\alpha_n) x_n$, which again
clearly belongs to $\text{Span}(S)$.

Note $-(\alpha_1 x_1)$
 $= -\alpha_1 x_1$

* Thus $\text{Span}(S)$ is an abelian group (Subgroup of V)
(Since V is abelian)

* For any $\alpha \in F$ and element $\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n \in \text{Span}(S)$
we see that $\alpha(\alpha_1 x_1 + \dots + \alpha_n x_n) = (\alpha \alpha_1) x_1 + \dots + (\alpha \alpha_n) x_n \in \text{Span}(S)$

Thus $\text{Span}(S)$ also has the scalar multiplication of V .

* (V1), (V2), (V3) and (V4) automatically holds as $\text{Span}(S) \subseteq V$.
Thus $\text{Span}(S)$ is a subspace of V .

H.W.:

① $\{0\}$, lines passing through origin and $\mathbb{R}^2$ are the
three kinds of subspaces of $\mathbb{R}^2$.

② Let α, β, γ be scalars with $\gamma \neq 0$.

$$\text{Let } W = \{(x, y, z) \in \mathbb{R}^3 : \alpha x + \beta y + \gamma z = 0\}$$

Prove that W is a Subspace of $\mathbb{R}^3$.

• W is called a plane in $\mathbb{R}^3$.

[Refer to earlier H.W.]

Linearly independent vectors

Let V be an F -vector space. Elements (or vectors) $v_1, v_2, \dots, v_n$ are said to be F -linearly independent if

any linear combination $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = 0$ implies $\alpha_1 = 0, \alpha_2 = 0, \dots, \alpha_n = 0$.

Examples

① $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \in \mathbb{R}^2$ are F -linearly independent

Proof If $\alpha_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0$ for some $\alpha_1, \alpha_2 \in F$ then we have

$$\begin{pmatrix} \alpha_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \alpha_2 \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = 0. \text{ That is, } \alpha_1 = 0 \text{ and } \alpha_2 = 0.$$

② Let $0 \neq X \in \mathbb{R}^2$ and Y be any point not on the line $\{\alpha X : \alpha \in \mathbb{R}\}$. Then X and Y are $\mathbb{R}$ -linearly independent.

Proof Let $A = \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix}$, where $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $Y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$.

We have already observed that A^{-1} exists and that $A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = X$, $A \begin{pmatrix} 0 \\ 1 \end{pmatrix} = Y$.

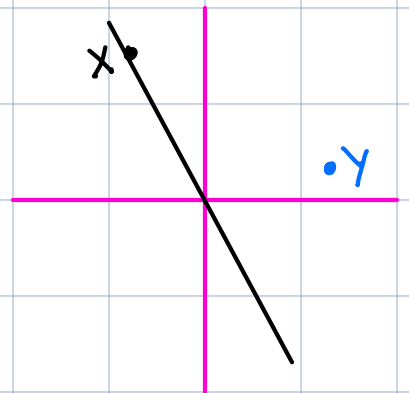
Suppose that $\alpha X + \beta Y = 0$.

$$\text{Then } A^{-1}(\alpha X + \beta Y) = A^{-1}0$$

$$\Rightarrow \alpha A^{-1}X + \beta A^{-1}Y = 0$$

$$\Rightarrow \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0.$$



Thus $\alpha = 0$ and $\beta = 0$.

③ $0 \neq v \in V$ then v is F -linearly independent. (why?)

Hint: $\alpha v = 0 \nleftrightarrow \alpha \neq 0$
 $\Rightarrow v = 0$.

Definition A nonempty subset $S \subseteq V$ is said to be F -linearly independent if every finitely many elements of S are F -linearly independent.

Definition. Elements $v_1, \dots, v_n \in V$ are said to be F -linearly dependent if they are not F -linearly independent

That is, there exists $\alpha_1, \alpha_2, \dots, \alpha_n \in F$ with at least one $\alpha_i \neq 0$ such that

$$\alpha_1 v_1 + \dots + \alpha_i v_i + \dots + \alpha_n v_n = 0$$

Examples:

① $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \end{pmatrix} \in \mathbb{R}^2$ are F -linearly dependent

$$0 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0 \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} + 1 \cdot \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0$$

Lin. Dep.

$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}; \alpha \neq 0$

$$-\alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + (-\beta) \begin{pmatrix} 0 \\ 1 \end{pmatrix} + 1 \cdot \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0$$

$$-2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + (-3) \begin{pmatrix} 0 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 2 \\ 3 \end{pmatrix} = 0$$

② If $0 \in S \subseteq V$ then S is F -linearly dependent

③ ④ Any two points on a line passing through origin are F -linearly dependent

$$x, y \in L = \{ \alpha x : \alpha \in F \}$$

$$\Rightarrow y = \alpha x$$

$$\Rightarrow 1 \cdot y + (-\alpha) x = 0.$$

⑤ Any three points on a plane passing through origin are F -linearly dependent. (Needs work!)

Remark: Suppose $v_1, v_2, \dots, v_m \in \mathbb{R}^n$ are F -linearly independent and A be an invertible $n \times n$ matrix. Then the vectors $Av_1, Av_2, \dots, Av_m$ are F -linearly independent.

Proof Suppose that

$$\alpha_1 Av_1 + \alpha_2 Av_2 + \dots + \alpha_m Av_m = 0$$

$$\text{Then } A(\alpha_1 v_1 + \dots + \alpha_m v_m) = 0$$

$$\Rightarrow \alpha_1 v_1 + \dots + \alpha_m v_m = A^{-1}0 = 0.$$

$$\Rightarrow \alpha_1 = 0, \alpha_2 = 0, \dots, \alpha_m = 0.$$

Thus

$Av_1, Av_2, \dots, Av_m$ are F -linearly independent.

